Arjun Parthasarathy

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EDUCATION

Columbia University — **New York, NY**

B.A Computer Science; B.A Applied Math — Expected May 2027

- **GPA:** 4.05 / 4.0
- **Honors:** I.I Rabi Scholar, Dean's List
- **Relevant Coursework:** Operating Systems, Computer Architecture, Compilers, Analysis of Algorithms, Partial Differential Equations, GenAI and Modern Deep Learning

University of Oxford — **Oxford, UK**

Visiting Student, Mathematics and Computer Science — Jan 2026 – Jun 2026

- **Relevant Coursework:** Numerical Linear Algebra, Topology, Rings and Modules, Number Theory, Calculus of Variations, Proof Systems

WORK EXPERIENCE

Jump Trading — **Chicago, IL**

Quantitative Research Intern – Jump Core Strategies — Jun 2026 – Sep 2026

- Training infrastructure - compiler optimization and autotuning pipelines for kernel fusion
- Custom TorchInductor lowering and CUTLASS kernels for Hopper and Blackwell GPUs

Department of Electrical Engineering, Columbia University — **New York, NY**

Research Assistant — Mar 2025 – Present

- Multi-token prediction using Transformer VAEs
- Gradient-based planning for model predictive control, co-advised by Yann LeCun (NYU)
- Advised by Micah Goldblum (Columbia)

Department of Electrical Engineering, Columbia University — **New York, NY**

Research Assistant — May 2025 – Jul 2025

- Writing energy-efficient and performant GPU kernels for dynamic voltage and frequency scaling
- Advised by Tanvir Ahmed Khan

HazyResearch, Stanford AI Lab, Stanford University — **Stanford, CA**

Research Assistant — Jul 2024 – Oct 2024

- Co-author of paper which received ICLR 2025 Spotlight Award
  - ThunderKittens: Implementing CUDA primitives for NVIDIA Hopper GPUs
- Designed FFT kernels for long convolution and added support for complex-number operations

SELECTED PUBLICATIONS

Closing the Train-Test Gap in World Models for Gradient-Based Planning
Workshop on World Models in Physical AI, NeurIPS 2026

ThunderKittens: Simple, Fast, and *Adorable* Kernels
ICLR 2025

SEIFER: Scalable Edge Inference for Deep Neural Networks
Workshop on Challenges in Deploy and Monitoring ML Systems, NeurIPS 2022

Partitioning and Placement of Deep Neural Networks on Distributed Edge Devices to Maximize Inference Throughput
32nd International Telecommunication Networks and Applications Conference (ITNAC) 2022

DEFER: Distributed Edge Inference for Deep Neural Networks
Workshop on Machine Intelligence in Networked Data and Systems, COMSNETS 2022